

The Role of the Conformational Profile of Polysaccharides on Skin Penetration: The Case of Hyaluronan and Its Sulfates

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The literature data suggest the capacity of biomacromolecules to permeate the human skin, even though such a transdermal permeation appears to be governed by physicochemical parameters which are significantly different compared to those ruling the skin permeation of small molecules. On these grounds, the present study was undertaken to investigate the *in vitro* diffusion properties through the human epidermis of hyaluronic acid and their sulfates. Low- and medium-molecular-weight hyaluronic acids and the corresponding derivatives at two degrees of sulfation were then tested. *In vitro* studies evidenced that the sulfated polymers permeate better than the corresponding hyaluronic acid, despite their vastly greater polarity, while the observed permeation markedly decreases when increasing the polymer's molecular weight regardless of the sulfation degree. Using a fluorescent-labeled polysaccharide, it was also evidenced that hyaluronans have a great affinity for corneocytes and likely cross the *stratum corneum* mainly through a transcellular route. The molecular-dynamics study revealed how the observed permeations for the investigated polysaccharides can be rationalized by monitoring their conformational profiles, since the permeation was found to be directly related to their capacity to assume extended and flexible conformations.

1. Introduction. – Transdermal administration is a well-established and attractive route of administration. However, the absorption of a drug applied on the skin is limited by the *stratum corneum*, which presents a multi-layered wall-like structure in which flat keratinized corneocytes are embedded in a lipophilic network of ceramides, cholesterol esters, and fatty acids. The barrier properties of this dense layer were investigated in-depth, revealing that lipophilicity and molecular size play key roles in determining the transdermal permeation of small molecules [1][2].

Conversely, the transdermal permeation of biological drugs was scantily investigated, even though it is commonly accepted that their passive diffusion is limited by both large molecular volume and ability to elicit H-bonds. However, polysaccharides have been successfully and widely applied through the skin, even though very little data on their permeation process have been hitherto reported in literature.

Thus, very few and often contrasting studies are reported for heparin [3], heparinoids [4], chondroitin [3], and hyaluronic acid (also called hyaluronan; HA). As an example, HA was reported to be unable to penetrate the *stratum corneum* in *in vitro* experiments due to its high hydrophilicity and stiffened random-coil molecular

configuration [5][6]. In another study performed *in vivo*, a gel loaded by 3% (w/w) radiolabelled HA applied to human forearm was able to penetrate into the skin reaching the dermis within 30 min [7].

Aiming at elucidation of the penetration mechanism of HA through human epidermis, the present work was focused on the impact of molecular weight, conformation, and polarity on the diffusion process of polysaccharides using low- and medium-molecular-weight HA and the corresponding derivatives at different degree of sulfation (HAS). Besides their relevant physicochemical profiles, these biomacromolecules were chosen for their significant impact in the biomedical field, since they are widely used as hydration agents, free-radical scavengers [8] and skin-penetration enhancers [9].

The transdermal penetration of HAs was experimentally determined by the *Franz* diffusion cell method using human epidermis as a membrane and confirmed by using confocal laser scanner microscopy (CLSM), while the possible relationship between the polymer's conformation and the ability to permeate the human epidermis was studied *in silico* by molecular-dynamics (MD) simulations. Indeed, despite its relevance in living organisms and artificial biosystems, the conformational properties of the HAs have been scantily investigated at an atomic level [10]. The hitherto reported computational studies have involved only very short oligomers [11], and the effects of sulfonation on the conformation of HA have never been analyzed by MD simulations.

2. Results and Discussion. – 2.1. *In vitro Penetration Studies.* Since the sensitivity of the analytical methods permitted to quantify permeants only 24 h after application (Table 1), it was possible to determine neither the flux of HAs through the skin nor the lag time.

Table 1 compiles the amounts of the considered HAs diffused across the human *stratum corneum* and epidermis as quantified in the receiving phase after 24 h and shows that molecular weight and sulfation degree strongly influence the permeation. Thus, an in-depth analysis of Table 1 allows two major considerations. First, the comparison of HAs having the same molecular weight but differing in the degree of substitution reveals that the *stratum corneum* and epidermis were more permeable to

Table 1. *The Main Characteristics of Hyaluronic Acids (HAs), the Corresponding Derivatives at Two Degrees of Sulfation, HAS, and Medium-Molecular-Weight HA Conjugated with Fluorescein Isothiocyanate, HAF. The relative amounts diffused across human stratum corneum and epidermis after 24 h (Q_{24}) of treatment are also reported (mean $\pm$ standard deviation, $n=3$).*

Name	Substitution degree [% mol]	M_r [kDa]	Polydispersity index	Q_{24} [nmol/cm ²]
HA	0	6.7	1.352	1.0 $\pm$ 0.6
		170.2	1.596	0.1 $\pm$ 0.0
HAS	1	6.8	1.259	4.9 $\pm$ 0.8
		181.3	1.483	0.6 $\pm$ 0.2
	2.5	6.4	1.452	2.4 $\pm$ 1.0
HAF	0	200.1	1.390	0.2 $\pm$ 0.1
		190.3	1.432	0.1 $\pm$ 0.0

the sulfated derivatives compared to unsubstituted HA. In detail, the reported permeated amounts evidence that the lower substitution degree (*i.e.*, 1% mol) shows favorable skin penetration characteristics providing the highest permeated amounts.

Second, the comparison of the three pairs of macromolecules with similar degrees of substitution but different molecular weights indicates that the permeated amounts of low-molecular-weight compounds are *ca.* tenfold higher than the corresponding medium-molecular-weight compounds. Notably, this effect is similarly observed in all compared pairs, thus indicating that the molecular-weight effect is not influenced by the degree of sulfation.

2.2. Confocal Laser Scanner Microscopy (CLSM). The penetration pattern of HAs through human *stratum corneum* and epidermis was then estimated using a medium-molecular-weight HA conjugated with fluorescein isothiocyanate (FITC) as a fluorescent probe, hereinafter termed HAF for the sake of brevity.

The confocal images of untreated *stratum corneum* and epidermis stained by propidium iodide (red signal) clearly shows corneocytes as well as the nuclei of dead cells of epidermis (*Fig. 1, a*). After 30 min, exposure to HAF, a green signal appeared in the *stratum corneum* (*Fig. 1, b*) and reached the equilibrium within 2 h (*Fig. 1, c*).

In *Fig. 2*, the *z*-stack projections are compared, namely, the longitudinal views of the epidermis, recorded in skin exposed to the HAF solution (*Fig. 2, a*) or a solution containing only FITC (*Fig. 2, b*). HAF shows a great affinity for keratins of the *stratum corneum*, since the overall structure of corneocytes presented a homogeneous green colorimetry, while it is hardly visible in the epidermis (*Fig. 2, a*). Conversely, the *z*-stack projections recorded in the skin samples exposed to the FITC solution evidence only a slightly superficial fluorescence (*Fig. 2, b*), remaining constant over time (data not shown).

These results are in agreement with the bioimaging for the transdermal delivery of FITC after topical administration to the dorsal skin of female hairless mice as reported by Yang and co-workers [12]. Both results imply that FITC alone is unable to bind any

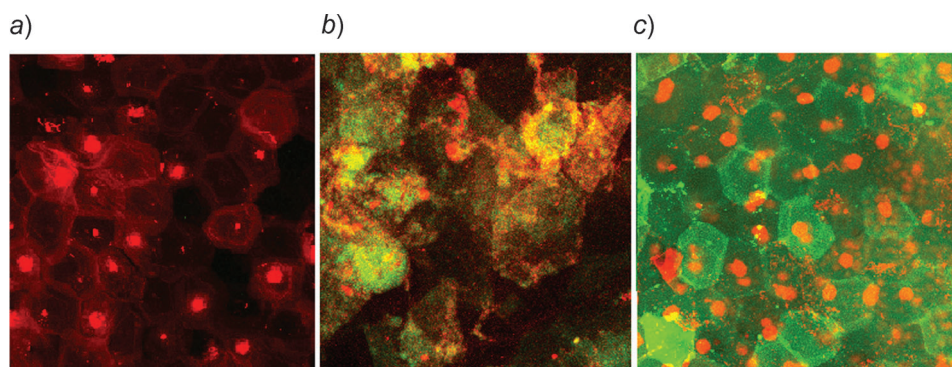


Fig. 1. CLSM Images showing distribution of HAF in human *stratum corneum* and epidermis with increasing durations of application: b) 30 min and c) 2 h. a) Accumulation of free propidium iodide in the inter-corneocyte spaces and the nuclei of dead cells of epidermis. Being lipophilic in nature, the dye is seen to accumulate in intercellular spaces (red stained). HAF appeared in the *stratum corneum* (green signal) 30 min after the application of the solution b) reaching the equilibrium in 2 h c).

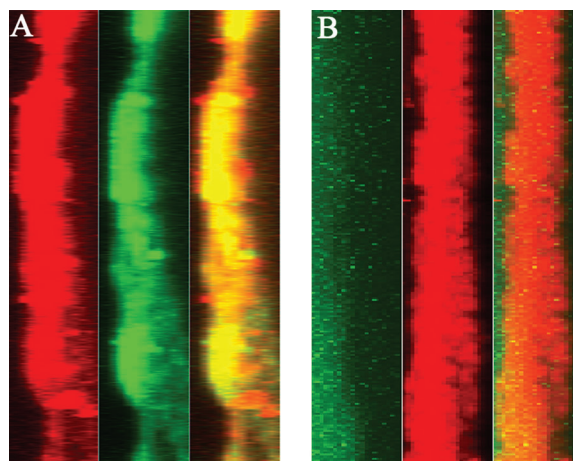


Fig. 2. CLSM Images (z -stack projections) showing human stratum corneum and epidermis 30 min after application of A) HAF and B) FTIC in aqueous solution. The green fluorescence corresponds to furrows where HAF and FTIC accumulated. The red color corresponds to the fluorescence emanating from the human stratum corneum and epidermis stained by propidium iodide and the yellow-green to HAF.

components of corneocytes, thus indicating that the fluorescent probe diffused through the stratum corneum and epidermis without being retained, unlike HA.

The qualitative results of confocal imaging are consistent with those obtained by gel-permeation chromatography. Indeed, FITC was quantified in the receptor phase ($0.02 \pm 0.01 \mu\text{g}/\text{cm}^2$) in the first sampling at 15 min, while HAF was detected only after 6 h and became quantifiable only after 24 h (Table 1). No other fluorescent signals were detected in the chromatograms indicating that HAF permeated without being significantly cleaved. This result is also consistent with literature data [7][12] and allows us to confirm that no significant degradation of HA occurred during the passive diffusion through stratum corneum and epidermis under our experimental conditions.

2.3. In silico Studies of Hyaluronans. To analyze the conformational profiles of the simulated HA chains, three descriptors were monitored: *i*) the radius of gyration which describes molecular size and shape, and is well-suited to examine the polymer flexibility [13]; *ii*) the end-to-end distance, namely the distance between one end of the polymer chain and the other end [14], and *iii*) the percentage of *synclinal* geometries in the torsions around the anomeric link (φ and ψ , as represented in Fig. 3). The first two descriptors are common indices used in polymer simulations, while the last parameter allows the overall structural properties to be linked to the local conformational changes in the glycosidic linkages.

The first part of the study compares the two non-sulfated polymers of different length (HA_{20} and HA_{10}). The reported values for the aforementioned descriptors (Table 2) show that the two polymers have a comparable conformational profile as suggested by a similar abundance of *synclinal* geometries in both torsions, but the longer HA_{20} chain shows a more compact structure (Fig. 4,a) as evidenced by both end-to-end distance and radius of gyration normalized per number of the repeating units.

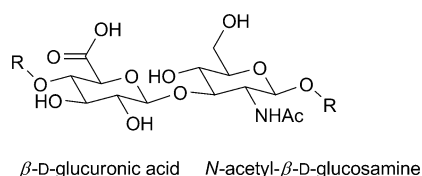


Fig. 3. Repeating unit of non-sulfated hyaluronic acid, i.e., ((1 $\rightarrow$ 4)- β -D-glucuronic acid-(1 $\rightarrow$ 3)-N-acetyl- β -D-glucosamine). Inter- and intrarepeating unit torsions ψ ϵ ϕ are also indicated.

Table 2. Average Values ($\pm$ standard deviation) for the Main Parameters Extracted from MD Runs for the Non-Sulfated HA Polymers of Different Length (HA₁₀ and HA₂₀) and the Longest Sulfated HA Polymer at Two Degrees of Sulfation (HA_{20S5} and HA_{20S10}). The conformations are defined as *synclinal* (in fact including also disfavoured *synperiplanar* geometries) when comprised in the $-90 \rightarrow +90$ ranges. In parenthesis, the average values of radius of gyration and end-to-end distance normalized per number of HA units are reported.

Oligomers	φ Synclinal geometries [%]	ψ Synclinal geometries [%]	Radius of gyration [$\text{\AA}$]	End-to-end distance [$\text{\AA}$]
HA ₁₀	60.0	100	10.32 ± 1.05 (1.03)	26.91 ± 4.34 (2.69)
HA ₂₀	61.5	100	15.67 ± 8.37 (0.78)	25.89 ± 8.57 (1.29)
HA _{20S5}	53.8	97.4	25.92 ± 1.43 (1.29)	59.62 ± 6.07 (2.98)
HA _{20S10}	64.1	97.8	21.51 ± 3.88 (1.08)	46.67 ± 4.15 (2.33)

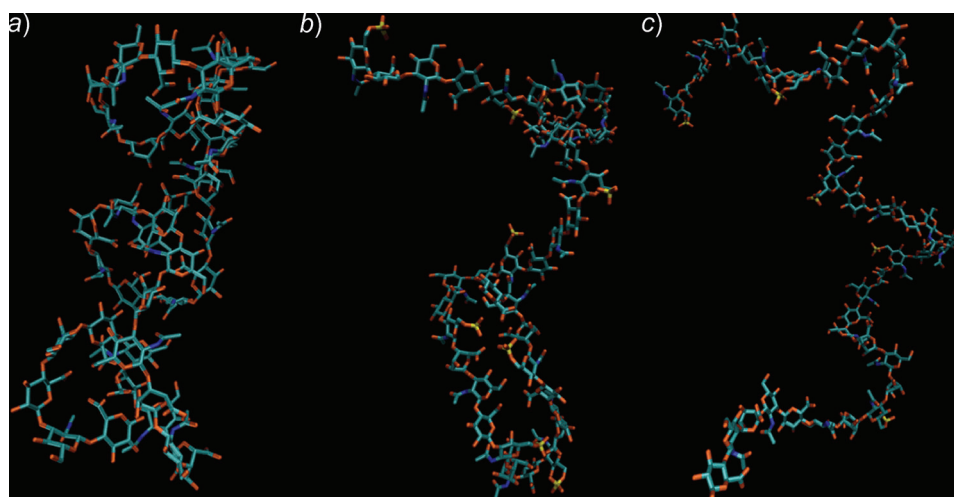


Fig. 4. Representative examples of the hyaluronan conformations as extracted by MD runs. a) Folded structure for non-sulfated HA₂₀; the helical structure is characterized by the axis of ca. 1.0 nm-a-loop diameter of ca. 1.2 nm and four disaccharides per turn. b) Intermediate conformation as obtained for HA_{20S10} which loses the helical structure, while showing a rather compact conformation in which the polymer ends fold on themselves forming elliptic structures whose shape and width should minimize the electrostatic repulsion between sulfonic groups. c) Extended and flexible structure as observed for HA_{20S5}.

These results confirm that, in aqueous solution and at low ionic strength, HA assumes a contracted left-handed helix conformation which is stabilized by an extensive H-bond network involving both solute–solvent interactions and inter-residue contacts such as those seen between the carboxylate of a glucuronic acid and the acetamide of the adjacent glucosamine [10]. Such a structure is in encouraging agreement with both NMR and X-ray data [15][16] and emphasizes the conformational versatility of the HA chains despite the caging effect of H₂O molecules around the glycosidic linkages.

The second part of the study involved the sulfated chains with a view to analyze how the sulfonation degree can influence the conformational behavior. Table 2 and Fig. 5 compare the longer non-sulfated HA₂₀ chain with the two differently sulfated chains, HA_{20S5} and HA_{20S10}, which can be seen as representative of the tested HAs with medium molecular weights, and with a sulfation degree of 1 and 2.5% mol, respectively. The performed MD simulations suggest that both sulfated polymers undergo large conformational changes and lose the helical architecture seen in the non-sulfated structures.

Although both sulfated chains show a substantial disruption of the helix motifs, Fig. 4 clearly shows the significant differences between them, because the less sulfated HA_{20S5} chain (Fig. 4, c) appears markedly more extended than the other one (Fig. 4, b). This is also documented by both end-to-end distance and radius of gyration, as well as by relative abundance of the *synclinal* geometries. Concerning the local conformation of the glycosidic linkages, the data in Table 2 confirm that the ψ angle assumes almost exclusively *synclinal* geometry in all performed simulations regardless of polymer length and sulfonation. In contrast, relevant differences are seen in the conformational profiles of the φ angle, since the percentage of *synclinal* geometries decreases in the HA_{20S5} polymer which appears characterized by more disordered and more extended conformations (*vide infra*), while the percentage in the HA_{20S10} chain remains comparable with the values observed for the non-sulfated polymers.

A detailed description of the conformational behavior of the simulated chains is provided by the radii of gyration (Fig. 5, a) and the end-to-end distances (Fig. 5, b).

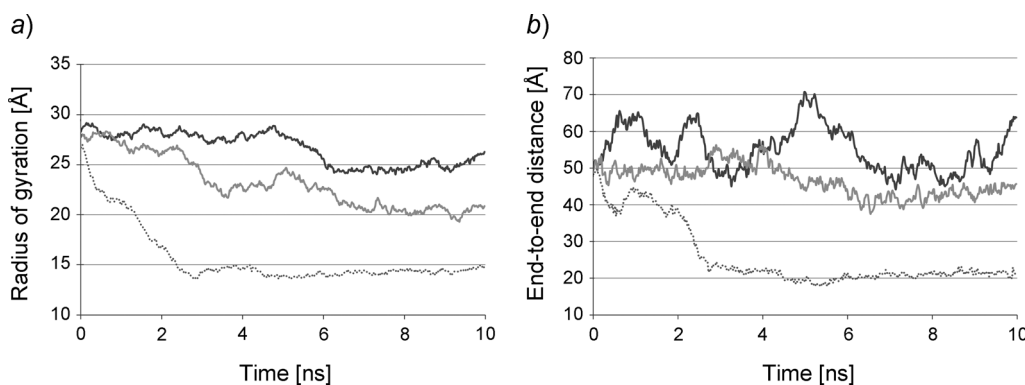


Fig. 5. Dynamic profiles of a) radius of gyration and b) end-to-end distance as computed by MD simulations of non-sulfated HA (HA₂₀, dashed grey line), HA₂₀ at low degree of sulfation (HA_{20S5}, black line), and HA₂₀ at high degree of sulfation (HA_{20S10}, grey line)

Both these descriptors indicate that the HA_{20S5} polymer assumes significantly extended conformations as confirmed by the highest values in both parameters, while the more sulfated HA_{20S10} chain presents an intermediate behavior characterized by disordered but more folded structures. Moreover, Fig. 5, b shows that the HA_{20S5} polymer has a vastly greater flexibility, since it repeatedly changes its whole conformation bouncing between conformations quite similar to those of HA_{20S10} (*i.e.*, $d \approx 50$ Å) and conformations vastly more extended (*i.e.*, $d \approx 70$ Å) without reaching an equilibrium phase.

Notably, Table 2 indicates that the described differences between sulfated and non-sulfated oligomers are maintained also when simulated in CHCl₃, which was chosen to simulate an apolar medium roughly comparable to the skin environment. Indeed, HA₂₀ assumes a contracted left-handed helix conformation also in CHCl₃, while HA_{20S5} conserves an elongated and disordered geometry. The differences between simulation in H₂O and CHCl₃ are, on average, marginal, even though one may note that the differences between sulfated and non-sulfated oligomers are a little reduced in CHCl₃, and this can be ascribed to the increased solvent weight which slows down the solute movements due to molecular friction as well as to the inability of the solvent to stabilize strong interactions with the simulated oligomers.

Altogether, the performed simulations allow some considerations which can be summarized as follows.

i) The non-sulfated chains assume quite ordered left-handed helices, and this is in clear agreement with the experimental data. As a trend, the simulations suggest that conformational compactness increases with polymer length.

ii) The sulfated chains lose the ordered helical motif and assume quite disordered structures, the less sulfated chains being more flexibly extended than the HA_{20S10} chain. In all simulated polymers, conformational rigidity increases with the number of charged groups, probably because the repulsion between them plays a dominant role with few charged groups, while an increased number of charged moieties tends to generate an ordered reticulum of ion pairs which stabilize the polymer conformation [17].

iii) The observed conformational differences were found to be not markedly influenced by the polarity of the simulated media. At least, they are somewhat affected by solvent friction which, however, cannot alter the overall conformational profile.

3. Conclusions. – Although the moisture content of *stratum corneum* and the aqueous composition of the epidermis permit a significant permeation also for hydrophilic compounds, the skin permeation of small organic molecules strongly decreases with increasing molecular size, polarity, and the number of H-bonding groups [1][18].

The present study clearly evidences that the investigated biopolymers do not obey the aforementioned physicochemical rules. Indeed, the reported experimental results indicated that HAs have a great affinity for keratin and likely cross human *stratum corneum* and epidermis mainly through corneocytes rather than intercellular lipidic network. Moreover, the obtained results clearly indicate that sulfated polymers penetrate to a larger extent than the corresponding HAs (Table 1), despite their vastly greater polarity.

The MD study can help to rationalize these data, shifting the focus to polymer flexibility and evidencing how the observed penetration behavior can be rationalized

by the polymer's conformational profile. In particular, the normalized values of radius of gyration were found to be in line with the extent of diffusion across the *stratum corneum* and epidermis over 24 h, the higher the values of radius of gyration, the higher the diffused amount.

Such a trend can be explained considering that the penetration increases with the capacity to assume extended and flexible structures which can easily adapt themselves to the skin environment, which deeply changes due to the different composition of the viable epidermis, corneocytes, and lipophilic network.

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Experimental Part

Chemicals. All tested hyaluronic acids (HAs) were kindly provided by *Fidia Farmaceutici* (Italy). As compiled in *Table 1*, the study involved six different HAs including both sulfated (HAS) and nonsulfated polymers differing in molecular weight (low molecular weight, *ca.* 6.5 kDa, and medium molecular weight, *ca.* 185 kDa) and in the sulfation degree (1 and 2.5% mol, resp.). The reported polydispersity indices (*Table 1*), which encode for the distribution of polymer's molecular mass and tend to unity for homogenous samples, confirm the satisfactory homogeneity of the tested polysaccharide samples. The study also involved a medium-molecular-weight polymer conjugated with fluorescein isothiocyanate (FITC), an aliquot of 1 mg/ml of which contained 0.2378 μg of fluorescent probe (*Fidia Farmaceutici*, Italy), hereinafter termed HAF for the sake of brevity. At the end of the depolymerization process, the medium-molecular-weight HA was dialyzed against H_2O by using a membrane with a 30 kDa molecular weight cut-off (*Centricon® Amicon*, Millipore, US). The absence of oligomers with molecular weights lower than 30 kDa in the dialyzed fractions was assessed by the carbazole method [19], which allows the quantification of D-glucuronic acid-free units produced from the HA hydrolysis. NaN_3 and propidium iodide (PI) soln. were purchased from *Sigma–Aldrich* (Italy). The H_2O used in the experiments was produced by a *MilliQ®* system (Millipore, US). All solvents were of anal. grade, unless specified.

In vitro Permeation Studies. The *stratum corneum* and epidermis used in the permeation studies was obtained from the abdominal skin of donors, who underwent cosmetic surgery (30–50-year-old, Eurasian female). The samples were prepared according to the internal standard procedure [20]. The full-thickness skin was sealed in evacuated plastic bags and frozen at -20° within 24 h after removal. Prior to experiments, the skin was thawed at r.t., and the excess of fat was carefully removed. The skin sections were cut into squares of *ca.* 2.5 cm^2 and, after immersing the skin in H_2O at 60° for 1 min, the epidermis was gently separated from the remaining tissue with forceps and carefully inspected by light microscopy for any defects.

Afterwards, the membrane was mounted on the *Franz* diffusion cell [20], whose receptor compartment (volume, 6.0 ml, exactly measured) was filled with degassed H_2O containing 100 $\mu\text{g}/\text{ml}$ NaN_3 as preservative. Special care was taken to avoid air bubbles between the buffer and the membrane in the receptor compartment. The upper and lower parts of the *Franz* cell were sealed with *Parafilm®* and fastened together by using a clamp, with the membrane acting as seal between the donor and receptor compartments. The system was kept at 37° with a circulating H_2O bath so that the membrane surface temp. was $32 \pm 1^\circ$ throughout the experiments. Before starting the experiments, the temp. at the surface of each receptor compartment was measured by using a digital thermometer (*PCE-T390*, *PCE Corporation*, Italy). The donor compartment was filled by 2% (*w/w*) aq. soln. (0.5 ml) and closed. At pre-determined times (1, 2, 4, 6, and 24 h), 0.2-ml samples was withdrawn from the receiver compartment and immediately replaced with fresh receiver medium. To avoid false positive results due to HA eventually present in the human skin, control experiments were carried out in parallel using H_2O as donor phase. The amount of HA eventually present in the receptor phase was subtracted by the sample reading.

A set of experiments was also carried out to determine the penetration mechanism of HAF into the human epidermis. After 0.5, 1, 2, 3, 4, and 24 h, the membranes were dismantled from the *Franz* cells, immediately washed with H₂O, and gently stripped by an adhesive tape strip (*Transpore tape*, 3M, US) to remove the unadsorbed permeant. Then, the samples were analyzed by confocal laser scanner microscopy (CLSM). A permeation experiment using a 5-mg/l aq. soln. of FITC as donor phase was also carried out. Sampling of receiving phase was carried out after 0.25, 0.5, 1, 2, 3, 4, and 24 h of application.

All reported values represent the averages of parallel experiments performed in triplicate.

Confocal Laser Scanner Microscopy (CLSM). The samples of gently stripped membranes were placed on a glass slide with the *stratum corneum* side up, stained by a PI soln. Acquisition of fluorescent images was performed with a *Zeiss LSM 510* confocal microscope (63× objective) with sequential acquisition setting at 1024×1024 pixels resolution. Each image was a z-series projection, with 1 μm interval between each plane. The absence of any cross-talk between HAF and PI fluorescence was ensured by a multi-track setting. Three to four sites for imaging were randomly selected in each coverslipped section.

Quant. Determination of HA. The amount of HAS was determined in terms of S content by inductively coupled plasma (ICP) MS analysis after depolymerization and mineralization. The contained S percentage was used to estimate the amount of permeant. The concentration of HA was measured by using the carbazole method [19]. In both cases, the permeated amounts were calculated by subtracting the amount of S or glucuronic acid determined in the blank.

HAF was determined by a size-exclusion chromatography (HPLC HP 1100 Chemstations) using a *BioSuite™* 125 column, 4 μm *UHR SEC*, 4.6×300 mm (*Waters*, US) eluted with 0.1M phosphate buffered saline (PBS) at pH adjusted to 8, at a flow rate of 0.35 ml/min and temp. of 25°. The eluent was monitored fluorometrically (excitation, 494 nm; emission, 520 nm; photomultiplier (PMT) gain, 10). Measurements were conducted with an injection volume of 100 μl and the retention time of HA was *ca.* 5.8 min. The end of the run was determined by injecting FITC (retention time, 17 min). The HAF and FTIC contents were determined by using a standard calibration curve ranging from 0.1 to 20 μg/ml and 0.01 to 5 μg/ml, resp.

In silico Construction of the HA Oligomers. The oligomers of HA were generated starting from the 3D structure determined by NMR (PDB Id: 2BVK) [16] which is composed by four repeating units (HA₄) of (1→4)-β-D-glucuronic acid-(1→3)-N-acetyl-β-D-glucosamine (as shown in *Fig. 3*) and assumes a contracted left-handed fourfold helix geometry. The base HA chain was then obtained by multiplying per five resolved starting structure to generate a molecule composed by 20 repeating units (HA₂₀). Two chains of different sulfonation degrees were built by modifying the base oligomer at the OH group at C(6) of the N-acetyl-β-D-glucosamine residue. Specifically, the two sulfated polymers were constructed by modifying one D-glucosamine residue every two (HA_{20S10}) and one D-glucosamine residues every four (HA_{20S5}). To investigate the effects of molecular length on HA folding, a non-sulfated chain composed by ten repeating units (HA₁₀) was also considered.

Overall, four HA polymers were simulated: two non-sulfated oligosaccharides of different length (HA₁₀ and HA₂₀) and two sulfated oligosaccharides with the same length but differing in the sulfonation degree, *i.e.*, HA_{20S10} and HA_{20S5}. Even though the differences in the simulated chains cannot exactly match the differences in the tested polysaccharides, due to the fact that short oligomers were here simulated, the four simulated polymers can reveal the conformational effect that a HA experiences with varying molecular weight and sulfation degree. The conformational profile of the generated polymers was analyzed by clustered MonteCarlo procedures (as implemented by VEGA suite of programs) [21] which produce 2,000 conformers by randomly rotating the flexible bonds pertaining the glycosidic linkages. The so obtained lowest-energy structures were finally minimized and used in the following MD simulations.

MD Simulations. Firstly, the polymers were neutralized by adding the necessary number of Na⁺ ions whose pose was computed by SODIUM program [22] and then inserted in a 70 Å×90 Å×70 Å box of H₂O molecules (apart from HA₁₀, which was solvated by a smaller 50 Å×50 Å×50 Å H₂O box). It should be noted that, since the initial structure of all simulated oligomers corresponds to a rather extended conformation, the considered simulation space should be large enough to allow a complete flexibility of the oligomers without biasing constraints. After a preliminary minimization to optimize the relative position of H₂O molecules and ions, the obtained systems underwent 10-ns MD simulations with

the following characteristics: *i*) Newton's equation was integrated using the r-RESPA method (every 4 fs for long-range electrostatic forces, 2 fs for short-range nonbonded forces, and 1 fs for bonded forces); *ii*) the simulation space was stabilized by introducing periodic boundary conditions (PBC; $75 \text{ \AA} \times 100 \text{ \AA} \times 75 \text{ \AA}$; $55 \text{ \AA} \times 55 \text{ \AA} \times 55 \text{ \AA}$ for HA₁₀); *iii*) the long-range electrostatic potential was treated by the Particle Mesh Ewald (PME) summation method ($60 \times 90 \times 60$ grid points; $50 \times 50 \times 50$ points for HA₁₀); *iv*) the temp. was maintained at $300 \pm 10 \text{ K}$ by means of the Langevin's algorithm *v*) Lennard–Jones (*L–J*) interactions were calculated with a cut-off of $10 \text{ \AA}$, and the pair list was updated every 20 iterations; *vi*) a frame was stored every 20 ps, to yield 500 frames. The simulations were organized in two phases: an initial period of heating from 0 to 300 K over 30,000 iterations (30 ps, *i.e.*, 10 K/ps) and the monitored phase of 10 ns. All MD runs were carried out using NAMD2.7 [23] with the force-field CHARMM v36 and Gasteiger's atomic charges on a 16-core Tyan VX50 system. All mentioned minimizations were performed using the conjugate gradient algorithm until a r.m.s. equal to $0.01 \text{ kcal mol}^{-1} \text{ \AA}^{-1}$.

Similarly, HA₂₀ and HA_{20S5} were simulated in a CHCl₃ cluster to analyze their conformational behavior in an apolar medium which brings to mind the skin's hydrophobic environment. The CHCl₃ cluster was built starting from a pair of interacting CHCl₃ molecules generated by semi-empirical calculations using MOPAC2010 and adopting the PM6 level of theory. The initial cubic box ($135 \text{ \AA} \times 135 \text{ \AA} \times 135 \text{ \AA}$; 14,075 molecules) was built by the solvent builder as included in VEGA program and starting from the pair of interacting CHCl₃ molecules. The obtained system underwent a preliminary minimization to optimize the relative position of solvent molecules, and then an equilibrating 1-ns MD simulation, which had the above-described characteristics. Such a CHCl₃ cluster was then utilized to solvate the HA₂₀ and HA_{20S10} oligomers, which underwent 10-ns MD runs applying the same computational protocol described above for the simulations in H₂O.

The conformational profile of the simulated polymers was analyzed by monitoring the two torsion angles, φ and ψ , which describe the glycosidic linkage between two sugar units, while the puckering of furanose rings was not considered. As depicted in Fig. 3, the φ angle pertains to the rotatable bond connecting the anomeric C-atom of *i*th saccharide with glycosidic O-atom, while the ψ angle refers to the rotatable bond connecting the glycosidic O-atom to the C-atom of *i* + 1th saccharide. Specifically, the φ torsion is the dihedral angle defined by the following atoms H_{ai}–C_{ai}–O–C_{i+1}, while the ψ torsion is defined by C_{ai}–O–C_{i+1}–H_{i+1}. For the sake of simplicity, the angle values were analyzed without discriminating between inter- or intrarepeating unit torsions. To do this, an *ad-hoc* script was written for VEGA software which automatically recognizes the described torsion angles and can also computes average values when analyzing trajectory files.

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